

Project Portfolio

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1. **Project:** Design and Simulation of a Hybrid Heating System for a University Campus Using a Modulating Groundwater-Source Heat Pump with R1234ze(E)

Objective: To thermodynamically design and simulate a hybrid heating system for a university campus using a modulating groundwater-source heat pump with R1234ze(E) refrigerant and a natural-gas boiler for peak-load coverage, with focus on heat-pump performance under varying outdoor temperatures and part-load conditions.

Achievements:

- Developed an Excel-based thermodynamic model of the complete heat-pump refrigeration cycle using CoolProp, including evaporator, compressor, condenser, superheating, and subcooling.
- Determined refrigerant state properties and visualized the heat-pump cycle using log p-h and T-s diagrams for R1234ze(E).
- Evaluated compressor isentropic efficiency, refrigerant mass flow rate, compression ratio, and electrical power demand, using compressor performance data from Danfoss Coolselector2 as a reference.
- Calculated COP and seasonal heat-pump performance under varying outdoor temperatures and part-load operation, accounting for changes in building heating demand and supply temperature through a modulating heating curve.
- Integrated the groundwater heat source, modulating heat pump, and auxiliary natural-gas boiler into a hybrid heating concept and analyzed the system against the university campus heating-load profile.
- Performed a basic design of the groundwater-side system, considering source temperature, required heat extraction, water properties, and heat-exchanger temperature differences.

Tools Used: Microsoft Excel, CoolProp, Danfoss Coolselector2

2. **Project:** Design of a Cross-sectoral Energy Efficient System for the District Heating for a Town Using Python Pandas and oemof.

Objective: To present the design, simulation, and cost-optimization of a cross-sectoral district heating and electricity supply system for a German town using Python and the Open Energy Modelling Framework (oemof).

Achievements:

- Designed a complete cross-sectoral district energy system integrating heat pumps, CHP units, boilers, solar PV, and energy storage.

- Sized thermal and electrical storage components based on realistic demand profile.
 - Developed and compared three operational scenarios, Calculated CAPEX and OPEX, LCOE, and CO₂ emissions.
 - Delivered results through technical visualizations and provided a recommendation based on sustainability and model performance.
 - Achieved system optimization by performing sensitivity analysis.
3. **Project:** Simulation of a Solar-Powered Organic Rankine Cycle (ORC) with R134a, R245fa, and Acetone as Working Fluids
- Objective:** To model and analyze a small-scale solar-powered Organic Rankine Cycle and evaluate the influence of different working fluids and operating conditions on thermodynamic performance, solar collector requirements, and overall electrical efficiency.

Achievements:

- Developed an Excel-based thermodynamic model of a solar-powered ORC using Dowtherm A as the heat-transfer fluid, incorporating the pump, evaporator/steam generator, expander, and condenser.
 - Calculated thermodynamic state properties, mass flow rates, turbine and pump work, heat addition/removal, cycle efficiency, global electric efficiency, and required solar mirror area.
 - Generated and analyzed T-s and t-Q diagrams to evaluate ORC processes, heat-exchanger performance, and pinch/approach temperature conditions.
 - Compared the performance characteristics of R134a, R245fa, and Acetone, including the influence of superheating, regeneration, and reheating on cycle efficiency.
4. **Master's Thesis**
- Title:** Transformation of Steelmaking Processes Through the Use of Residual Materials
- Institution:** Fraunhofer UMSICHT, Sulzbach-Rosenberg, Germany
- Objective:** To investigate how residual biogenic materials (e.g., waste wood, shredder light fraction, green waste) can be integrated into steelmaking processes to reduce greenhouse gas emissions and dependency on fossil fuels, with a focus on enhancing the SALCOS® green steelmaking route for Salzgitter AG.

Key Research Areas:

- **Technology Assessment:** Evaluated state-of-the-art biomass conversion technologies (e.g., pyrolysis, HTC) and shortlisted technology according to project requirements.
- **GHG Analysis:** Conducted Life Cycle Assessment (LCA) to quantify GHG emissions for various biomass integration scenarios.
- **Hydrogen Production:** Estimated hydrogen production from Pyrolysis and Gasification pathways. Modeled hydrogen generation from biochar gasification using Aspen Plus.

- **Biomass Integration:** Assessed suitability of biochar and syngas for substitution in Electric Arc Furnace (EAF) and Direct Reduction Processes (DRP) based on feedstock availability.
- **Strategic Impact:** Proposed a novel multi-feedstock, cross-sectoral pathway for green steel production and carbon reduction via bioenergy inputs.

Methodologies & Tools Used:

- **Software:** Aspen Plus, Microsoft Excel
- **Simulation Models:** Equilibrium-based gasification model
- **Frameworks:** ISO 14040/44/67 for LCA and GHG accounting
- **Field Work:** Direct communication with technology providers and literature review of commercial thermochemical conversion technologies

5. **Project:** Experimental Investigation of Hydraulic Injection Characteristics of Alternative Sustainable Fuels

Institution: Kompetenzzentrum für Kraft-Wärme-Kopplung (KWK), Amberg, Germany

Objective: To experimentally analyze and compare the injection behavior of alternative fuels Hydrogenated Vegetable Oil (HVO), Dimethyl Carbonate blend (DMC20), and Diesel (B0) in a modern diesel common rail injection system.

Achievements:

- Used a professional-grade injector test bench to measure fuel injection parameters like: Injection rate (mg/ms), Injection duration (ms), and Trigger signal (A).
- Analyzed performance of three fuels under varying injection pressures (750 and 1000 bar) and fuel quantities (5–70 mg).
- Used test bench experimental data (.ASC files), plotted results to visualize performance differences, and identified anomalies.

6. **Project:** Design of a Farm with the Cooling and Drying of Agriculture Products by Using Solar Photovoltaics and Bioenergy.

Objective: To design a renewable-powered farm, integrating solar photovoltaics and biogas to meet the energy needs of agricultural and dairy operations with a focus on sustainability, reliability, and economic viability.

Achievements:

- Estimated and analyzed energy demand profiles for core operations (milk cooling, maize drying, lighting, water pumping, and ventilation).
- Designed and sized a solar PV system to meet the demand of self-sufficiency using Polysun software. Evaluated off-grid feasibility, identifying seasonal energy deficits and proposing battery integration and demand-side management solutions.
- Calculated biogas requirements and dryer energy loads based on maize yield, fan power, and moisture content. Mirroring real-world farm engineering practices.

- Conducted economic assessments and suggested system improvements such as CHP integration, and solar thermal heating.

7. **Project:** Process Simulation of Hydrogen Production by Glycerol Steam Reforming

Objective: To simulate and compare multiple process configurations for hydrogen production from glycerol steam reforming using Aspen Plus, with the goal of identifying optimal feed ratios and reformer conditions for maximizing H₂ output and process efficiency.

Achievements:

- Developed and validated two process models (stoichiometric and equilibrium-based) for steam reforming of glycerol in Aspen Plus.
- Conducted sensitivity analysis to optimized H₂ production output by varying feed ratios and reformer temperatures, identifying an ideal water-to-glycerol ratio and thermal range.
- Implemented heat integration techniques to reduce reactor duty, improve energy efficiency and proposed system modification.

8. **Presentation Title:** Carbon Capture and Storage (CCS): Ecological & Economic Perspectives

Objective: To assess the environmental and economic viability of CCS technologies in achieving net-zero targets, with focus on current methods, challenges, and global deployment status.

Achievements:

- Explained major CCS pathways and scientific laws governing the absorption process principles (Henry's Law), CO₂ partial pressure, and how these affect energy and cost efficiency.
- Assessed current global CCS deployment, highlighting the gap between installed and needed capacity to meet 2030 net-zero targets.
- Addressed cost drivers, ecological risks, land use, CO₂ utilization (CCU), and direct air capture (DAC).

9. **Lab work:** Photovoltaic System Evaluation and Performance Testing

Key tasks conducted:

- Measured and analyzed I-V and P-V characteristics of monocrystalline and amorphous solar modules under varying irradiance and temperature conditions.
- Conducted performance experiments on module tilt angle, azimuth orientation, and their effect on current output and power.
- Connected solar modules in series and parallel configurations and studied their voltage-current behavior with and without shading and bypass diodes.
- Performed real-world data analysis on roof-mounted PV systems installed at university using 24-hour measurement datasets (south-facing and east-west systems).

10. **Scientific Poster:** Sustainability in Shared Transport: A Systematic Review
(Scientific Poster Presentation & Extended Abstract)

Institution: OTH Amberg-Weiden, Amberg, Germany

Objective: To investigate the barriers faced by older adults and people with disabilities in accessing shared public transport and propose actionable, inclusive solutions through a systematic literature review.

11. **Presentation Title:** Effects of Parking Management on Urban Mobility

Objective: To explore how strategic parking policies influence urban mobility, space efficiency, and sustainable transport behavior in high-density cities.

Key tasks conducted:

- Reviewed international urban parking regulations and their impact on public transport accessibility and car dependency.
- Analyzed smart parking systems using case studies from Cologne and Zurich.
- Evaluated policy tools such as parking pricing, permit systems, and maximum space requirements.
- Created and delivered a structured presentation summarizing challenges, data-driven insights, and policy recommendations.